

PAPER_927: GW190425 Phonon-Suppressed Gravitational Wave Strain

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Abstract

Derives the UQFF phonon-suppressed gravitational wave strain for the GW190425 binary neutron star merger: $h_{\text{UQFF}}(t) = h_{\text{GR}}(t) * D_{\text{total}} * \exp([SSq]t/26)$, where $D_{\text{total}} = 0.530$ represents a 47% strain damping via phonon-vacuum coupling at 1.25 THz. The suppression factor arises from the 26-layer polylog structure redistributing gravitational wave energy into phonon modes. Time evolution through $\exp([SSq]t/26)$ provides dynamical strain modification during the inspiral. GW190425's mass-gap nature ($M_{\text{chirp}} \sim 1.44 M_{\text{sun}}$) makes it the ideal testbed for phonon effects at the NS-BH boundary.

1. Core Equations

Section A: Lagrangian

$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) * D_{\text{total}} * \exp([SSq] * t / 26)$
 $D_{\text{total}} = 0.530$ (UQFF phonon suppression factor)
 $h_{\text{GR}} = 3.0\text{e-}22$ (GR strain at $d_L \sim 159$ Mpc)
 $[SSq] = 0.57$ (calibrated coupling constant)

Section B: VDS/DVP/BH Number Systems

VDS: $D_{\text{total}} = \text{Product}_{\{k=1\}^{\{26\}} (1 - [SSq]*\Phi_k/(26*\Phi_0))$
DVP: $h_n = h_{\text{GR}} * \text{Product}_{\{k=1\}^{\{n\}} D_k$ (layer-by-layer suppression)
BH: $\Lambda_{\text{tidal}}^{\text{phonon}} = \Lambda_{\text{GR}} * (1 + \Phi*S_{26}*[SSq]/N)$

Section SM: SM Anchors

$M_1 = 1.7 M_{\text{sun}}$, $M_2 = 1.5 M_{\text{sun}}$ (GW190425 component masses)
 $M_{\text{chirp}} = (M_1*M_2)^{(3/5)} / (M_1+M_2)^{(1/5)} \sim 1.44 M_{\text{sun}}$
 $d_L \sim 159$ Mpc (luminosity distance)
 $f_{\text{GW}} = 20\text{-}300$ Hz (LIGO band)

2. Parameters

Parameter	Default	Description
h_{GR}	3.0e-22	GR strain amplitude
D_{total}	0.530	Suppression factor
t	0 s	Observation time
f_{GW}	300 Hz	GW frequency

3. Key Results

Configuration	h_{UQFF}	Suppression
$D_{\text{total}} = 0.530$	$1.59\text{e-}22$	47.0%
$D_{\text{total}} = 0.333$	$1.00\text{e-}22$	66.7%
$D_{\text{total}} = 0.667$	$2.00\text{e-}22$	33.3%

4. Physical Interpretation

The 47% strain suppression for GW190425 implies that nearly half the gravitational wave energy is redistributed into phonon vacuum modes during propagation through the 26-layer UQFF structure. This is consistent with the GW170817 result (66.7% suppression, PAPER_916) when accounting for the different chirp masses and distances. The mass-gap nature of GW190425 (total mass $3.2 M_{\text{sun}}$, above typical NS-NS but below confident BH formation) may reflect phonon-mediated stability at the NS-BH transition, where vacuum buoyancy provides additional pressure support.

5. References

- PAPER_916: GW190425 mass-gap phonon suppression (Session 210b)
- PAPER_917: Exponential strain phonon evolution (Session 210c)
- ns_phonon_gw190425_wstp.py: 5-class standalone module
- WSTP expression #27: $h_{\text{UQFF}}(t)$ GW190425